



Oceanographic regimes in the northwest Labrador Sea since Marine Isotope Stage 3 based on dinocyst and stable isotope proxy records



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ABSTRACT

Sea surface temperature (SST), salinity and density gradients in the upper water column of the northwest Labrador Sea have been reconstructed based on high resolution analysis of a core (HU2008-029-004PC) spanning the last ~36 ka, raised off Hudson Strait. The modern analogue technique was applied to dinocyst assemblages and combined with stable isotope data from *Neogloboquadrina pachyderma* left-coiled (Npl) for this purpose. Three oceanographic regimes were identified, broadly corresponding to the “glacial”, “deglacial” and “postglacial” intervals. The site remained under the direct influence of the Laurentide Ice Sheet (LIS) margin until the postglacial and did not record the Bølling-Allerød warming and weakly recorded the Younger Dryas event. The “glacial” regime lasted until ~12.2 cal ka BP. It was characterized by generally low concentrations of dinocysts within an assemblage indicative of quasi-perennial sea ice. The “deglacial” regime (ca 12.2–8.3 cal ka BP) was marked by increased biogenic fluxes and more diversified dinocyst assemblages and possibly an enhanced subsurface inflow of North East Atlantic Deep Water. Warm summer (~11 °C) but low winter (~0 °C) sea surface temperatures, sea ice cover during about 3 months per year, and low summer salinity (~28) suggest strong stratification in the upper water layer in relation to meltwater supply from the LIS. Following the final drainage of glacial Lake Agassiz through Hudson Strait, which is dated here at ~8.3 cal ka BP, and the subsequent LIS collapse, increased summer salinity (up to ~35) was accompanied by a reduced seasonal gradient of sea surface temperature from winter (~3.8 °C) to summer (~8.6 °C) suggesting enhanced penetration of North Atlantic Water. Weakened stratification of the surface water layer then allowed for winter convection and Labrador Sea Water formation, which is consistent with increased Npl- $\delta^{13}\text{C}$ values in response to higher ventilation of the subsurface water layer.

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1. Introduction

Labrador Sea Water (LSW) is an important component of the modern Atlantic Meridional Overturning Circulation (AMOC). The cold winter surface air temperatures above the Labrador Sea cool a western branch of the saline North Atlantic Drift, which sinks due to increased density and forms the intermediate LSW (Lazier, 1973) and contributes to North Atlantic Deep Water (NADW) (Yashayaev and Loder, 2009 and references therein). The rate of LSW formation is variable and responds to both changes in atmospheric circulation and salinity, which affect stratification and therefore convective mixing. As such, periods of warmer climate with increased freshwater flux can inhibit LSW formation. Over one third of Arctic freshwater export presently flows via the Canadian Arctic Archipelago and Nares Strait into the Labrador Sea (Serreze et al., 2006),

making Hudson Strait and Davis Strait very important Arctic freshwater pathways (Curry et al., 2011). Therefore any change in temperature and freshwater outflow from the Arctic may modify the LSW production rates as during the Great Salinity Anomaly (e.g., Gelderloos et al., 2012). It may also modify the strength of the AMOC as suggested from modeling experiments (Goosse et al., 1997; Wadley and Bigg, 2002; Cheng and Rhines, 2004). In this context, it is relevant to investigate the past history of water masses and sea ice formation, in relation to freshwater fluxes through the northern Labrador Sea to quantify the potential impacts of future freshwater fluxes and increasing sea surface temperature on the formation of LSW.

Throughout the last glacial cycle, the Labrador Sea has been marked by meltwater supplies from the northeastern margin of the Laurentide Ice Sheet (LIS). Sedimentological and paleoceanographic studies of marine cores from the shelf and slope off Labrador and eastern Baffin Island have permitted to identify phases of LIS margin advance, waning, retreat, and Heinrich Events (Hillaire-Marcel et al., 1994; Andrews et al., 1994b, 1998, 2012; Jennings

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et al., 1998; Hillaire-Marcel and Bilodeau, 2000; Rashid et al., 2003; Rashid and Piper, 2007). However, the impact of changing climate and meltwater pulses on the oceanic conditions (temperature, salinity, sea ice cover, upper water mass stratification) has yet to be determined on a regional scale, especially in the northwest Labrador Sea and at the outlet of Davis Strait and Hudson Strait.

For this study, we had access to a new core (HU-2008-029-004) ideally located in the northern Labrador Sea. The core was recovered from mid-slope, about 200 nautical miles east of Hudson Strait and south of Davis Strait. It spans over 36 ka and provides detailed information on the impacts of meltwater pulses from the north-eastern margin of the LIS on the regional ocean conditions and northwest North Atlantic circulation. Sea surface temperature (SST), sea surface salinity (SSS), seasonal cover of sea ice, and productivity (gC m^{-2}) were reconstructed from dinocyst assemblages (de Vernal et al., 2001, 2005, 2008, 2013; Radi and de Vernal, 2008). Subsurface dwelling planktonic foraminifera were analyzed for oxygen and carbon stable isotopes to provide complementary information about subsurface temperature and salinity, ventilation, variations in convective mixing and intermediate water formation (e.g., Ravelo and Hillaire-Marcel, 2007), and also information on major meltwater pulses and their impact on sea ice production rates (Hillaire-Marcel and de Vernal, 2008).

2. Modern hydrographic setting and location of core collection

The hydrography of the Labrador Sea is influenced by both relatively warm and saline waters flowing from the south and cold low-salinity waters flowing from the north (Fig. 1). Along the western Greenland margins, the West Greenland Current (WGC), which flows to the north, consists of a mix of the cool and low saline waters (temperature $\sim -1.8^\circ\text{C}$, salinity ≤ 34.5) carried through the East Greenland Current (EGC) along the shelf. The warm, saline North Atlantic waters (core at 200–700 m, temperature $\sim 4.5^\circ\text{C}$, salinity ≤ 34.95) that are transported via a western branch of the Irminger Current (IC) flow above the slope (Cuny et al., 2002). Along the Labrador Shelf, the Labrador Current (LC) flows southward. It is formed from the Baffin Island Current (BIC) (temperature $\sim -1.5^\circ\text{C}$, salinity ≤ 34) and outflow from the

Hudson Strait, which both consist of cold and low saline Arctic waters. Along the upper slope, the LC overlies Irminger water that has circulated around Baffin Bay. Below the surface layer, the Labrador Sea Water (LSW; temperature $\sim 3.0^\circ\text{C}$, salinity ~ 34.9) is formed by vertical convection due to the sinking of dense waters cooled in winter (Lazier, 1973). LSW has depths reaching down to 2500 m above the North East Atlantic Deep Water (NEADW) and the Denmark Strait Overflow Water (DSOW) (Yashayaev, 2007).

Piston core HU-2008-029-004 (henceforth PC04) was collected in 2008 off the Southern Baffin Island shelf in the northern Labrador Sea (61.46°N , 58.04°W ; Campbell et al., 2009) at a water depth of 2163 m (Fig. 1). At the coring site, summer surface waters are predominantly those carried by the WGC (Wu and Tang, 2011). The present mean winter and summer SSTs are $3.7 \pm 0.5^\circ\text{C}$ and $7.3 \pm 1.2^\circ\text{C}$, respectively, and summer SSS averages 34.2 ± 0.3 (NODC 2001). Sea ice is occasional and occurs only once every 3 years on average. The 1953 to 2003 sea ice compilation using data provided by the National Snow and Ice Data Center (NSIDC) in Boulder CO indicates that the core site has a mean of 0.7 ± 1.3 months per year with more than 50% of sea ice, calculated with observations for 1–5 months per year for 15 years.

3. Methods

The core PC04 is 896 cm in length. Onboard measurements included magnetic susceptibility, spectrophotometry, and sedimentary descriptions, which can be found in the HU2008029 cruise report (Campbell et al., 2009) together with core photographs. The core was not showing any handling disturbance. The working half was sampled onboard at 1-cm intervals. For this study, subsamples were taken at every 4 cm throughout the core, for a total of 220 samples.

The chronostratigraphy of PC04 was established from radiocarbon dates of planktonic foraminiferal populations that consist of >95% *Neogloboquadrina pachyderma* left-coiled (Npl). Accelerator mass spectrometry (AMS) radiocarbon measurements were made at Lawrence Livermore National Laboratory and at the National Ocean Sciences AMS Facility of Woods Hole. Radiocarbon ages were calculated using the Libby half-life of 5568 years and normalized to a $\delta^{13}\text{C}$ of -25‰ . The ages were converted to calibrated years and modeled using the Marine09 calibration curve (Reimer et al., 2009) with a marine reservoir correction of 400 years in OxCal 4.2 (Ramsey, 2008). No additional correction (ΔR) was made as it was the case for other ^{14}C -based chronologies from planktonic foraminifera in the area (see a discussion in Hillaire-Marcel et al., 2007; supplementary material). We have chosen to use the *p_{sequence}* model in OxCal, which relies on a Bayesian approach using information that includes the ^{14}C dates, their depths, and the changes in deposition rate for modeling the probability of each age (Ramsey, 2008). This approach was regarded as adequate considering the changes in sedimentary processes related to the dynamics of the LIS margin during glaciation and deglaciation, including Heinrich Events. The resulting age model assumes constant ^{14}C -carrier flux settling through the water column to the bottom, constant habitat (water depth) within and among Npl samples, and homogeneous mixed layer at the sediment–water interface (Berger and Johnson, 1978; Bard, 2001). The calibrated ages are reported as the modeled median cal. years BP (see Table 1).

For palynological preparations, 5 cm³ subsamples were rinsed and sieved through 106 μm and 10 μm sieves. The dried >106 μm size fraction, which mostly contains detrital material, was weighed as a proxy for ice rafting deposition. It was also used to hand pick foraminifera for stable isotope analyses. The 10–106 μm size fraction was processed following the detailed methods described by de Vernal et al. (1999). The residual organic matter was mounted onto a slide with Kaiser's glycerol gelatin. Palynological analyses were

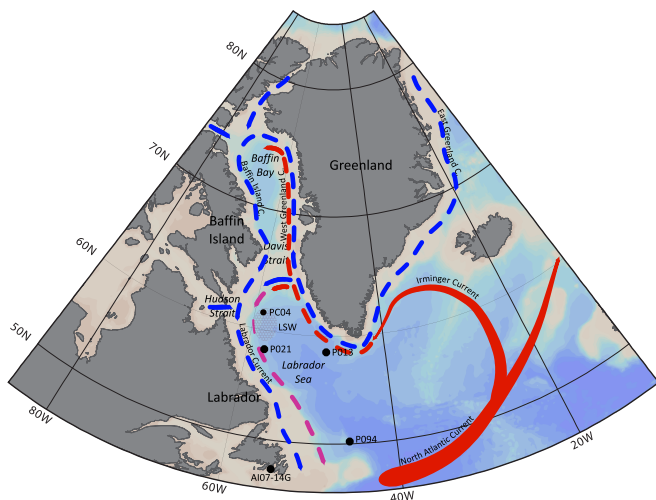


Fig. 1. Map indicating PC04 coring location (895 cm long piston core; 61.46°N , 58.04°W ; 2163 m water depth), with surface currents (North Atlantic Current (NAC), Irminger Current (IC), East Greenland Current (EGC), West Greenland Current (WGC), Baffin Island Current (BIC), and Labrador Current (LC)). Neighboring cores P013, P021, P094 and AI07-14G are also displayed. Labrador Sea Water (LSW) is formed in the Labrador Sea approximately at the location indicated by the hatched oval.

Table 1
Radiocarbon dates from planktonic foraminifers (*Neogloboquadrina pachyderma* left-coiled).

Depth (cm)	CAMS/NOSAMS (OS) Laboratory number ^a	AMS- ¹⁴ C ages (years BP) ^b	Calibrated age interval (2σ) (cal years BP) ^c	Modeled age interval (2σ) (cal years BP) ^d	Modeled age median (cal years BP)
20–21	CAMS-153491	2370 ± 60	2151–1841	2159–1828	1998
28–29	OS-96931	3880 ± 25	3953–3750	3953–3751	3850
36–37*	CAMS-155459	7845 ± 30	8416–8181	8418–8179	8319
68–69*	CAMS-155461	7975 ± 35	8416–8181	n/a	n/a
84–85*	CAMS-155462	7735 ± 30	8416–8181	8441–8196	8335
96–97	CAMS-153492	8780 ± 60	9544–9302	9540–9300	9447
116–117	CAMS-155463	9290 ± 30	10,218–10,033	10,218–10,009	10,147
216–217	CAMS-153493	10,990 ± 45	12,630–12,380	12,620–12,375	12,473
244–245	CAMS-153494	11,400 ± 120	13,128–12,636	13,071–12,635	12,816
388–389	CAMS-153495	12,700 ± 60	14,878–13,952	14,871–14,010	14,363
488–489	CAMS-153496	13,360 ± 60	16,207–15,076	16,359–15,191	15,734
592–593	CAMS-153497	15,120 ± 60	18,445–17,615	18,447–17,614	17,898
704–705	CAMS-153498	20,860 ± 170	24,922–23,932	24,907–23,928	24,407
764–765	OS-96934	21,600 ± 160	25,906–24,903	25,903–24,919	25,353
816–817	CAMS-153499	25,400 ± 370	30,687–29,105	30,655–29,005	29,848

^a The analyses were made at Lawrence Livermore National Laboratory (CAMS) or the National Ocean Sciences AMS Facility (NOSAMS).

^b Radiocarbon age was calculated using the Libby half-life of 5568 years and corrected with a $\delta^{13}\text{C}$ of -25‰ . *Three ¹⁴C ages were averaged (7852 ± 55) prior to calibration due to their association with the rapid sediment accumulation during the final drainage of glacial lake Agassiz some 8.4 cal ka ago (e.g. Barber et al., 1999; Hillaire-Marcel et al., 2007).

^c A marine reservoir correction of 400 years was applied with no additional correction (ΔR), and the ages converted to calibrated years using Oxcal 4.2 (Ramsey, 2008) and the Marine09 (Reimer et al., 2009) calibration curve.

^d The ages were modeled using Oxcal 4.2 (Ramsey, 2008). Modeled ages marked n/a indicate the interval was not used in the age model.

conducted by identifying and counting all palynomorphs, which include cysts of dinoflagellates (dinocysts), pollen and spores, organic linings of benthic foraminifera and reworked pre-Quaternary palynomorphs. The marker-grain method (Matthews, 1969) was used to calculate the number of total palynomorphs providing an accuracy of approximately $\pm 10\%$ for a 95% confidence interval (de Vernal et al., 1987). A minimum of 300 dinocysts were identified when possible at species or genera level and counted for the application of the modern analogue technique (MAT) to reconstruct sea surface conditions including temperature, salinity, sea ice cover, and productivity (e.g., de Vernal et al., 2008). Dinocyst species were identified following the nomenclature provided by Rochon et al. (1999), de Vernal et al. (2001), and Head et al. (2001). The pollen and spores were identified with reference to McAndrews et al. (1973) and Moore et al. (1991) and counted to evaluate the inputs from the terrestrial vegetation. Reworked palynomorphs include pollen, spores, dinocysts and acritarchs of pre-Quaternary age, which were differentiated from recent palynomorphs based on their taxonomic identity and preservation state. They are used as indicator of erosion of Phanerozoic rocks and subsequent outwash deposition. In the Labrador Sea, they have been used to identify detrital events (cf. Hiscott et al., 2001). Foraminifer organic linings are related to benthic productivity and can help determine the degree of calcium carbonate dissolution (de Vernal et al., 1992). The total numbers of dinocysts, pollen and spores, foraminifer linings, and total reworked palynomorphs are expressed as individuals per cm^3 of sediment. Relative abundances of dinocysts used for paleoceanographical reconstructions were calculated from the sum of dinocysts counted. A concentration threshold of 100 cysts cm^{-3} was used as minimum reliability requirement (Forcino, 2012).

The modern analogue technique (MAT) was used to reconstruct sea surface temperature (SST) in summer and winter, sea surface salinity (SSS) in summer, sea ice cover as expressed as the number of months per year with a concentration $> 50\%$, and productivity (gC m^{-2}). MAT was applied on the updated “modern” dinocyst database of the Northern Hemisphere that includes 1492 sites and 66 taxa (see database at <http://www.geotop.ca>; de Vernal et al., 2013), using the scripts prepared by Guiot (CEREGE, France) for the software R (<http://cran.r-project.org/>) and the procedure described by de Vernal et al. (2005). We applied a logarithmic

transformation to emphasize the weight of accompanying taxa. We calculated the most probable conditions from the average of the 5 best analogues weighted inversely to their distance, and the variance provided the upper and lower limits. Validation tests were performed by dividing the database in 5, with 1193 samples in the reference data set and 299 samples in the verification data set to evaluate the error or root mean square error of prediction (RMSEP), which also corresponds to the standard deviation of the difference between observation and reconstruction. The RMSEP established at ± 1.4 months/year for the sea ice cover, ± 1.2 °C and ± 1.6 °C for winter and summer SSTs and ± 2.6 for SSS.

The stable isotope compositions of oxygen ($\delta^{18}\text{O}$) and carbon ($\delta^{13}\text{C}$) were measured for the planktonic foraminifer *N. pachyderma* left-coiled (Npl), which is a mesopelagic species that lives in cold saline waters along the pycnocline and commonly found in Arctic waters (e.g., Bé and Tolderlund, 1971; Carstens et al., 1997; Simstich et al., 2003). Npl was chosen because it is the species that is by far the most abundant among planktonic foraminifera throughout the core. Actually, Npl constitutes an almost monospecific assemblage, which is expected given the location of the core in the Arctic province of the modern planktonic foraminifer distribution (cf. Kucera, 2007). Npl specimens were picked from the 150–250 μm size fraction of the > 106 μm residue from the palynological preparations. Stable isotopic analyses were conducted at GEOTOP by acidification with 102% orthophosphoric acid (Wendeborg et al., 2011) in a Multicarb™ preparation device coupled to an IsoPrime™ isotope ratio mass spectrometer. The isotopic composition for each sample was measured relative to the international reference, the Vienna Pee Dee Belemnite (VPDB), using the conventional “ δ -per mil” notation. Reference (NBS 19, IAEA) and working (UQ6 carbonate, GEOTOP, Hillaire-Marcel et al., 2004) standards were used for each analytical run. The reproducibility of measurements as estimated from daily measurements of the UQ6 standard material is better than $\pm 0.05\text{‰}$ at $\pm 1\sigma$ level for both isotopes (Hillaire-Marcel et al., 2004).

Some sedimentological parameters were used in conjunction with the micropaleontological proxies to provide better constraints on the sedimentary influx when creating the age model. These parameters included magnetic susceptibility, weight percent coarse fraction, and weight percent carbonate. The magnetic susceptibility data was collected shipboard with a Multi-Sensor Core Logger

(MSCL; cf. Campbell et al., 2009). The coarse fraction percentage was calculated using the dry weight of the $>106\ \mu\text{m}$ fraction relative to the initial dry weight of the sample, as indicated above. The low abundance of light biological remains in this fraction makes this measurement essentially representative of ice-rafted deposition. The weight percent carbonate was calculated from the inorganic carbon fraction ($\% C_{\text{inorg}}$) as the difference between the total carbon ($\% C_{\text{tot}}$) and organic carbon ($\% C_{\text{org}}$) as measured with a combustion furnace coupled to an elemental analyser (Carlo-Erba™). The C_{org} percentage was determined after acidification by fumigation for 24 h with 12 M HCl to remove the inorganic carbon and corrected for weight loss (Hélie, 2009). The C_{org} and C_{inorg} are estimates of the total organic and total inorganic contents respectively. However, potential biases may occur with detrital dolomite and/or HCl-leachable mineral and compounds present in the sediment (e.g., Maccali et al., 2013). XRD analyses indicate the presence of dolomite in the carbonate rich layers, resulting in minor differences of estimates for C_{inorg} contents (Laurence Nuttin, unpublished data).

4. Results

4.1. Chronostratigraphy

Lithology, magnetic susceptibility, weight % coarse fraction and weight % carbonate were used to gather prior information for constructing the age–depth model in OxCal, in addition to the 15 radiocarbon dates (Table 1). Sediment core PC04 consists of hemipelagic dark gray silty clay interbedded with two to three layers of graded mud intervals with laminae and sand and gravels (Fig. 2; cf. Campbell et al., 2009). These layers have low magnetic susceptibility

values, and high weight % carbonate and coarse fraction content (Fig. 2). They correspond to Labrador Sea detrital carbonate events as described previously by Andrews and Tedesco (1992), Andrews et al. (1994b, 1995) and Stoner et al. (1995, 1996). The sediment structure and coarse particles visible by CAT-scan images further suggest that these layers correspond to nepheloid-flow deposits as described by Hesse and Khodabakhsh (1998) and Rashid et al. (2003) from the Hudson Strait ice margin. The upper two layers are bracketed by ^{14}C dates (Table 1), which are within error of Heinrich Event (HE) intervals H1 and H2 in cores from the Labrador Sea (cf. Andrews and Tedesco, 1992; Andrews et al., 1994b; Hillaire-Marcel et al., 1994; Rashid et al., 2003; Stoner et al., 1995, 1996). Another detrital carbonate event probably corresponds to H3 (Stoner et al., 1996; Rashid et al., 2003), but it will not be considered further in this study due to poor age constraints. Since these layers are rapidly deposited (1–2 ka), the upper and lower stratigraphic boundaries of H1 and H2, as identified as the increase in carbonate content, have been used as stratigraphic boundaries in the model.

The ^{14}C samples at 36.5, 68.5, and 84.5 cm yielded almost identical dates. They indicate very rapid sediment accumulation rates at about $7852 \pm 55\ ^{14}\text{C}$ yrs BP as one may calculate from the average of the three dates. Since the 85–36 cm interval represents a short lived sedimentary phase, the three AMS dates were combined and two age–depth curves were made with OxCal, one for the lower part of the core below 85 cm and one for the upper part of the core above 36 cm (Table 1; Fig. 2). The modeled results provide an age range of 8441–8179 cal ka BP for the 85–36 cm interval, with an average value of 8327 cal ka BP. Such an age coincides with the final drainage of Lake Agassiz through Hudson Strait, as recorded in many cores from the Labrador margins (Barber et al., 1999; Hillaire-Marcel et al., 2007; Lewis et al., 2012). The age–depth model also

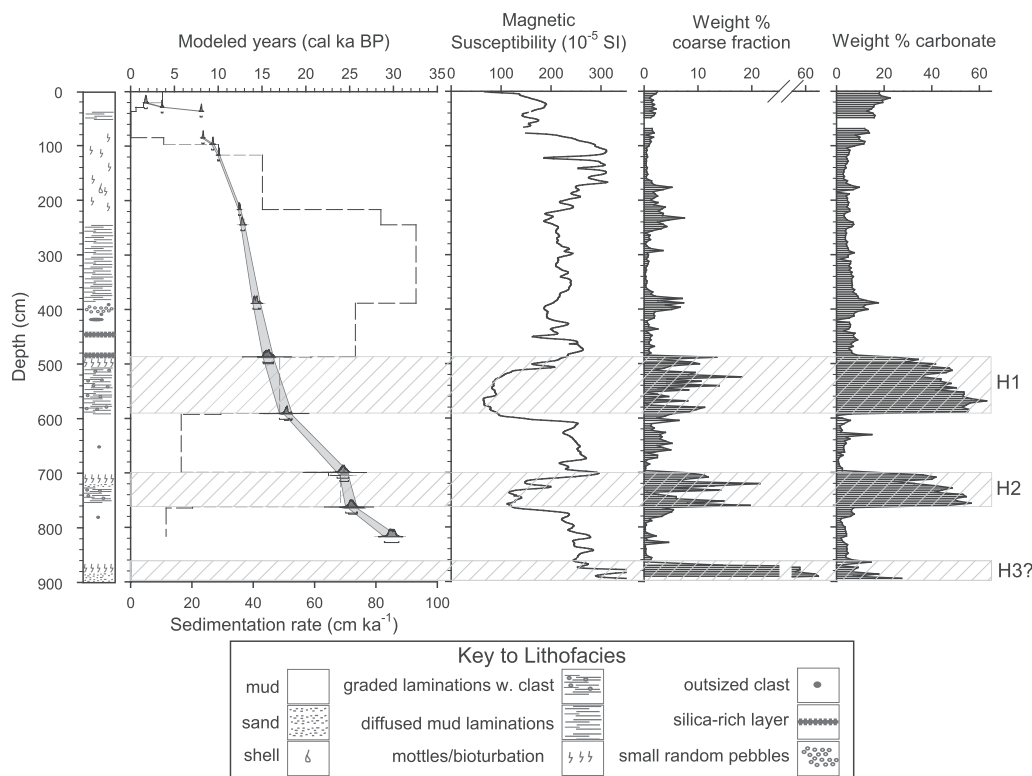


Fig. 2. Age vs. depth relationship for core PC04 based on lithology, magnetic susceptibility, weight % coarse fraction, weight % carbonate, and ^{14}C dates listed in Table 1. These parameters assisted in the establishment of Heinrich Events H1, H2 and possibly H3 identified as the hatched intervals. The rapidly deposited H1 and H2 provided prior information for the age model in the form of boundaries (solid horizontal lines). Two separate models were produced before and after the interval of rapid accumulation during the drainage of Lake Agassiz (e.g., Barber et al., 1999; Hillaire-Marcel et al., 2007). The sedimentation rates are represented by the dashed lines.

bracketed Heinrich Events 1 and 2 with calibrated ages of 17.9 and 15.7 cal ka BP, and 25.4 and 24.4 cal ka BP, respectively.

Linear interpolation between each modeled date and boundary was used to calculate the sedimentation rates and the age–depth relationship, which were used to plot proxy data against age.

4.2. Dinocysts and other palynomorphs

The dinocyst concentrations are relatively low, of the order of 10^4 cysts cm^{-3} . The minimum values are recorded during the glacial interval and until ~ 15.7 cal ka BP (<100 cysts cm^{-3} ; Fig. 3). There are a few intervals with slightly higher concentrations (from 100 to 1000 cysts cm^{-3}) centered at ~ 32 , 29, 24 cal ka BP. From ~ 15.7 cal ka BP, dinocyst concentrations increased with maximums of up to 6000 cysts cm^{-3} . Pollen and spore concentrations are low (Fig. 3), which is expected due to distance from the coast (Rochon and de Vernal, 1994) and the then reduced vegetational cover inland. Samples are mostly barren until ~ 15.7 cal ka BP, and increase to a maximum of 180 cm^{-3} grains and spores during the Holocene. Concentrations of foraminifer linings are highly variable within a 0 – 10^4 cm^{-3} range (Fig. 3). Moderate concentrations are present until after H2, decrease to <10 cm^{-3} until after H1, then increase to maximum values through the Holocene. These three groups of microfossils follow similar trends, with lower concentrations during the glacial interval and higher throughout the postglacial. The deglacial interval depicts moderate concentrations of these palynomorphs, possibly due to higher sedimentation rates.

Reworked palynomorphs include pollen, spores and dinocysts, in addition to acritarchs. They reflect erosion from the Paleozoic–Mesozoic sedimentary formations of the Canadian Arctic and West Greenland (e.g., Hiscott et al., 2001). They are present until ~ 8.3 cal ka BP (Fig. 3) with concentrations reaching as much as 1200 cm^{-3} . The reworked palynomorph peaks are correlated with H3(?), 2, and 1. They likely correspond to sedimentary reworking during triggered ice-surfing events.

The dinocyst assemblages are dominated by *Brigantedinium* spp., *Operculodinium centrocarpum*, *Nematosphaeropsis labyrinthus*, cysts of *Pentaparthodinium dalei* and *Islandinium minutum* (Fig. 4). They are accompanied by *Spiniferites elongatus*, *Spiniferites ramosus*,

Impagidinium sphaericum, *Impagidinium pallidum*, and *Selenopemphix quanta*. The complete dataset is available at <http://www.geotop.ca>.

The interval from ca 36 to 15.7 cal ka BP consists of nearly 100% *Brigantedinium* spp. From ca 15.7 to 12.2 cal ka BP the dinocyst assemblage (Fig. 4) consists of $>80\%$ *Brigantedinium* spp., and up to 12% *I. minutum* and 9% *S. quanta*.

After ~ 12.2 cal ka BP, the dinocyst assemblages are characterized by important changes. Several species emerge including *O. centrocarpum*, *N. labyrinthus*, the cysts of *P. dalei*, *S. elongatus*, *S. ramosus* and *I. sphaericum* (Fig. 4). *Brigantedinium* spp. decreases to less than 25% and *I. minutum* increases to a maximum of 30%. The relative abundance of taxa depicts important changes from ca 12.2 to 8.3 cal ka BP. In particular, there is an increasing trend of the cyst of *P. dalei*, which reached maximum values of 44% at ~ 8.3 cal ka BP.

The samples included in the ~ 8.3 cal ka BP-drainage layer may consist of a mixture of sediments from different environments (upper slope/shelf) and/or earlier deposits. Nonetheless, the dinocyst samples within this layer were analyzed assuming that they would mostly relate to productivity in the overlying water column, contemporaneous of the drainage event. The species assemblages of these samples are very similar to one another and comparable to underlying samples. However some contain $\sim 1\%$ of *Spiniferites mirabilis-hyperacanthus*, which is a temperate taxon (cf. Rochon et al., 1999).

From ca 8.3 to 2.0 cal ka BP, the occurrence of *I. minutum* decreases significantly and the assemblages are dominated by *O. centrocarpum* ($>60\%$). A near disappearance of *P. dalei*, a decrease in *Spiniferites* species, and the inception of *I. pallidum* are also observed. After ~ 2.0 cal ka BP, the relative abundance of *O. centrocarpum* decreases from >60 to 40% and *N. labyrinthus*, which is a species currently found in the subarctic waters of the Labrador Sea (Rochon and de Vernal, 1994; Rochon et al., 1999), increases from 20 to 60%.

4.3. Reconstruction of sea surface conditions

The reconstruction of sea surface conditions using MAT shows the succession of three very different types of environment (Fig. 5a). The northwest Labrador Sea was perennially to quasi-perennially ice covered throughout the glacial interval and until ~ 12.2 cal ka BP. Most samples from ca 36.6 to 15.7 cal ka BP contain extremely low dinocyst concentrations. The cyst counts are low and are therefore statistically weak for quantitative reconstruction. Nevertheless, the few intervals that yielded reliable assemblages (at ~ 32 , 29, 24 cal ka BP) led to reconstruct an almost perennial sea ice cover (>9 months per year), with low summer SSTs (<2.5 °C), low SSS (30–32.5) and extremely low productivity (<100 gC m^{-2}). We are thus confident that the intervals almost barren in dinocysts correspond to even harsher conditions, close to nil productivity and perennial sea ice cover. The intervals of increased concentration and higher species diversity recorded between ca 15.7–12.2 cal ka BP correspond to slightly milder conditions with sea ice for about 9 months per year and summer SST and SSS of about 1.7 °C and 31.4, respectively.

A dramatic shift occurred at ~ 12.2 cal ka BP with increased summer SSTs up to 11 °C (Fig. 5a, b). From ~ 12.2 to 8.3 cal ka BP, winter temperatures remained cold (0 °C), which resulted in large seasonal gradients of temperature. This interval is also characterized by particularly low salinity, which reached minimum values of 28, and by seasonal sea ice cover to an average of 3 months per year and by an increase in productivity to >300 gC m^{-2} . Whereas these conditions characterized the area until ~ 8.3 cal ka BP, some variability was recorded. For example, between ca 10.2 and 10.6 cal ka BP, winter and summer SSTs were at their highest (2.4

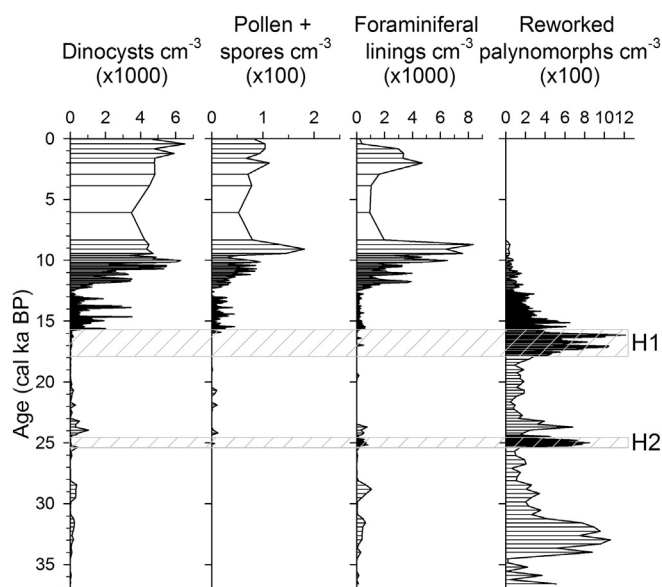


Fig. 3. Concentrations of dinocysts, pollen and spores, foraminifer linings, and reworked palynomorphs expressed in number of specimen per cm^3 of sediment for the last $\sim 36,600$ years in core PC04. Peaks of reworked palynomorphs are temporally correlated to Heinrich Events H1 and H2 as identified as the hatched intervals.

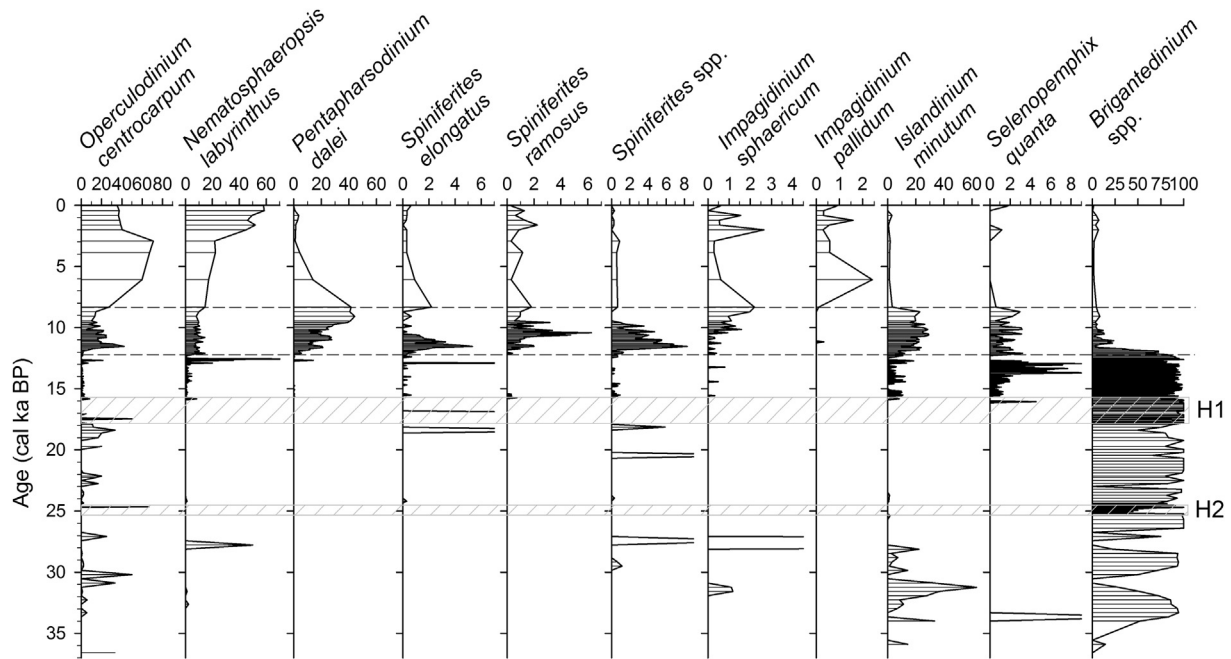


Fig. 4. Relative abundance (percentage) of the main dinocyst taxa in core PC04 for the last ~36,600 years (note the variable percentage scales). The complete assemblage and tabulation can be found on the GEOTOP website (<http://www.geotop.ca>). Ecostratigraphic units are separated by a dashed line. Heinrich Events H1 and H2 are identified as the hatched intervals.

and 11.4 °C respectively), and SSS and sea ice cover were at their lowest (29.2 and 1.6 months per year respectively).

After ~8.3 cal ka BP until present, the sea surface conditions fluctuate around modern values. Winter SSTs ranged from 1.4 to 3.8 °C, summer from 4.5 to 8.6 °C, salinity from 33.3 to 34.9, sea ice cover between 0 and 2.9 months per year, and productivity between 133 and 210 gC m⁻². Unfortunately the low temporal resolution of the Holocene prevents any assessment of variations in sea surface conditions that could correlate to climate/ocean trends depicted elsewhere in the Labrador Sea (e.g., de Vernal and Hillaire-Marcel, 2006).

4.4. Foraminiferal $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$

The stable oxygen and carbon isotopic analyses of 150–250 μm Npl were performed on 115 of the 220 intervals sampled (Fig. 5a). Many samples produced few Npl which made it impossible to perform representative isotopic analyses. $\delta^{18}\text{O}$ values increase from 4.0‰ to 4.8‰, from core-bottom to the LGM. Light excursions are recorded at the beginning and/or termination of Heinrich Events H2 (3.4‰) and H1 (3.3 and 2.3‰). Following H1, $\delta^{18}\text{O}$ values jump to ~4‰ then decrease gradually to <3‰ following the 8.3 cal ka BP event, and remain within a 2.6–3‰ range throughout the remaining part of the Holocene.

Large fluctuations are also observed in the $\delta^{13}\text{C}$ Npl record (Fig. 5a). The $\delta^{13}\text{C}$ values varied between 0.4‰ and –0.2‰, with a minimum value bracketing H1. A lesser pronounced $\delta^{13}\text{C}$ minimum also occurs between ca 11 and 12 cal ka BP (Fig. 5a, b). Then $\delta^{13}\text{C}$ values increased through the Holocene to a maximum of 1.0‰ by ~3 ka, followed by a decrease towards ~0.5‰ in near surface samples.

5. Discussion

Aside short events and high frequency fluctuations in sea surface conditions, the palynomorph concentrations, dinocyst assemblages, sea surface reconstructions and stable isotopes of Npl

highlight two major oceanographic changes over the last ~36 ka (Fig. 5a). They define three ecostratigraphic units that indicate the succession of different hydrographic environments in the area: (1) a glacial phase spanning from the base of the core to ca 12.2 cal ka BP, (2) a regional deglacial phase from ca 12.2 to 8.3 cal ka BP, and (3) a fully “postglacial” interval from ~8.3 cal ka BP to present. These ecostratigraphic units are discussed below with reference to large-scale oceanic circulation in the Labrador Sea and the glacial paleogeography of eastern Canada.

5.1. The glacial phase (ca 36.6–12.2 cal ka BP)

The glacial phase is characterized by extremely low concentrations of modern palynomorphs and foraminifera, indicating low productivity likely due to perennial or quasi-perennial sea ice cover (Fig. 3). Slightly higher dinocyst concentrations are observed, notably between ca 15.7 and 12.2 cal ka BP, with assemblages almost exclusively dominated by heterotrophic species such as *Brigantedinium* and *Islandinium* (Fig. 4) that tolerate harsh conditions with dense ice cover (de Vernal et al., 1997, 2001, 2013; Rochon et al., 1999). Reconstructions based on MAT confirm that modern analogues are from the Canadian Arctic where very cold conditions and quasi-perennial sea ice cover prevail (Fig. 5a). Therefore the northwest Labrador Sea was mainly perennially ice covered until 15.7 cal ka BP when conditions improved slightly allowing for some seasonal breakup and primary productivity. Sea surface reconstructions in sites located further south along the Labrador slope (P021; Fig. 1), also suggested harsh conditions during this interval (de Vernal et al., 2001). However, in the southern Labrador Sea (P094; Fig. 1), several high amplitude fluctuations of sea surface conditions with episodes of slightly warmer conditions and lesser extent of seasonal ice cover were reported (de Vernal et al., 2000, 2005). Such episodes marked by milder conditions were thus likely restricted to the southern Labrador Sea and did not extend to the northwest.

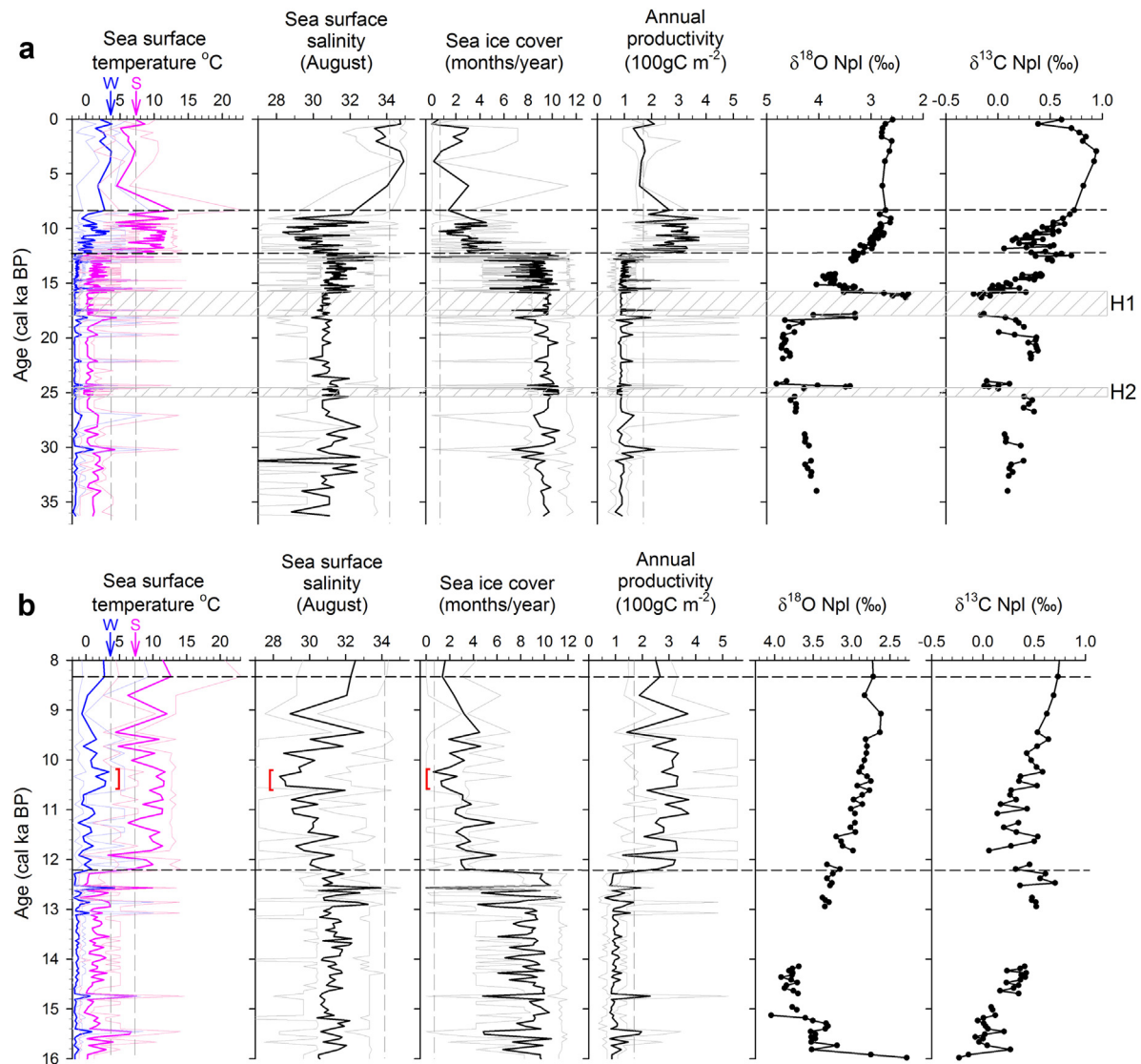


Fig. 5. Reconstruction of sea surface conditions from dinocyst assemblages and the oxygen and carbon stable isotopes of *Neogloboquadrina pachyderma* left-coiled (Npl) of core PC04. Sea surface temperatures (SST) in winter (w) and summer (s) are represented by blue and pink curves respectively. The lighter blue and pink curves correspond to maximum and minimum SSTs possible as calculated from a set of 5 modern analogues. Sea surface salinity, sea ice cover and annual productivity are represented by black (most probable values) and gray lines (minimum and maximum possible). Modern values of winter and summer sea surface temperature, sea surface salinity and sea ice cover, as provided by the NODC and NSIDC, are indicated by the vertical dashed lines. Ecostratigraphic units are separated by a horizontal dashed line. Heinrich Events H1 and H2 are identified as the hatched intervals. a. Data covering the past ~36,600 years. b. Blow up for the interval spanning 16–8 ka. Red brackets identify the ~10.4 cal ka BP extreme climate event (see text). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The high Npl- $\delta^{18}\text{O}$ values of the glacial interval are interrupted by light- $\delta^{18}\text{O}$ excursions at the end of H1 and H2 (Fig. 5a). Low concentrations of foraminifera, likely due to low productivity and dilution in sediment because of high depositional rates during HEs (Fig. 2), prevented analyses to be made within the detrital layers related to the events. In any case, the light excursions associated with HEs have been recorded in other cores of the Labrador Sea and interpreted as reflecting dilution by meltwater pulses (e.g., Hillaire-Marcel et al., 1994; Andrews et al., 1994a; Hillaire-Marcel and Bilodeau, 2000; Rashid et al., 2011). Hillaire-Marcel and de Vernal (2008) suggested that they do not exclusively relate to meltwater, but reflect the transfer of the low sea surface salinity signal to subsurface waters through the production of isotopically-light brines which sank to Npl-habitat depth, i.e., on the pycnocline between the surface and intermediate water layers. Since Npl requires growth in waters with salinity above ~34 and can tolerate

much higher salinities (Hilbrecht, 1996; Bergami et al., 2009), and since MAT reconstructions indicate continuously cold, low saline and densely sea ice covered surface waters, it seems likely that the terminations of the HEs were characterized by heavy ice production, causing the brine induced $\delta^{18}\text{O}$ shifts.

The Bølling-Allerød (BA; ~14.7–12.9 cal ka BP) and the Younger Dryas (YD; ~12.9–11.6 cal ka BP) are not obvious in the northwest Labrador Sea record. The sea surface reconstructions and Npl- $\delta^{18}\text{O}$ values show no clear indications for warmer conditions corresponding to the BA. Consequently, there is no evidence for cooling related to the YD (Fig. 5b). From around 14 to 13 cal ka BP, though, increased concentrations of dinocysts (Fig. 3) coupled with the highest calculated sedimentation rates in the core (Fig. 2) might suggest enhanced productivity. This interval could thus represent slightly improved conditions during the BA. Very low palynomorph concentrations (Fig. 3) coupled with a peak in coarse fraction

content during a brief episode between ~ 12.8 and ~ 12.5 cal ka BP (245–216 cm; Fig. 2) suggest higher detrital input likely related to ice-rafting activity, which might thus constitute the local signature of the early YD. However, there is no evidence for a significant change in sea surface conditions prior to ~ 12.2 cal ka BP. The high Npl- $\delta^{13}\text{C}$ values prior to ~ 12.2 cal ka BP (Fig. 5b) could indicate an enhanced ventilation rate of the subsurface water mass (e.g., Ravelo and Hillaire-Marcel, 2007). However the dense sea ice cover would inhibit vertical convection. Veum et al. (1992) and Bauch et al. (2001) also observed an increase in Npl- $\delta^{13}\text{C}$ in the GIN Seas during the YD, matching a similar trend in benthic- $\delta^{13}\text{C}$. They considered it as the result of intermediate water formation caused by vertical convection during the YD cooling. We are thus tempted to assign the ~ 12.4 – 12.2 ka $\delta^{13}\text{C}$ -peak in PC04 with the inflow of an intermediate component of the North Atlantic Deep Water produced in the GIN Seas during an interval marked by dense sea ice cover and strong stratification between the cold and diluted surface water layer in the Labrador Sea and warmer North Atlantic water. The $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ profiles in core PC04 are comparable to those from the more thoroughly dated P094 at Orphan Knoll (Fig. 1; Clarke et al., 1999; de Vernal and Hillaire-Marcel, 2000). However unlike P094, PC04 and other adjacent cores (Rashid et al., 2011) do not have a carbonate peak defining HO making it difficult to estimate the timing of the YD (e.g., Pearce et al., 2013) and the shift in oceanic regime.

5.2. The deglaciation phase (ca 12.2–8.3 cal ka BP)

Dinocyst species associated with the phototrophic productivity in subpolar-temperate waters, such as *Spiniferites* spp., *Pentapharsodinium dalei*, *Nematosphaeropsis labyrinthus* and *Impagidinium sphaericum* (e.g., Rochon et al., 1999), occurred in significant numbers at about 12.2 cal ka BP (Fig. 4), thus marking the end of full glacial conditions in the northwest Labrador Sea. A similar shift at the glacial–interglacial transition has been reported from the Labrador slope core P021 (de Vernal et al., 2001). Therefore this transition seems to be regionally consistent, and its timing, around 12.2 cal ka BP is well constrained from the record of core PC04.

The dinocyst assemblages of the 12.2–8.3 ka interval contain species characteristic of temperate waters (cf. above), but also taxa typical of seasonal sea ice cover such as *Islandinium minutum*. Therefore, the overall assemblages suggest strong seasonal gradients of temperatures. Accordingly, MAT reconstructions indicate that surface water conditions dramatically changed at 12.2 ka from very cold and quasi-perennial sea ice covered, to cool, fresher, and ice free for most of the year (Fig. 5b). The reduced salinity is associated with meltwater fluxes from the LIS (e.g., Dyke, 2004). It was accompanied by strong stratification of the upper water masses and resulted in low thermal inertia of the surface layer, which is consistent with large seasonal gradients of temperature in the surface layer marked by warm summers and freezing conditions in winter (cf. also Solignac et al., 2004).

During the deglaciation phase, the $\delta^{13}\text{C}$ values in Npl remained light until the final phase of the deglaciation (Fig. 5b). This indicates a lack of ventilation of the subsurface water mass (e.g., Ravelo and Hillaire-Marcel, 2007). The isotopic and dinocyst data together illustrate highly stratified upper water masses in the Labrador Sea during the deglaciation as previously shown by de Vernal and Hillaire-Marcel (2006). This stratification prevented vertical convection and intermediate water formation at a regional scale.

Although the sea surface conditions reconstructed from the PC04 dinocyst record fluctuated slightly throughout the deglacial phase, there is one prominent interval around 10.4 cal ka BP. MAT reconstructions are marked by minimum salinity values coupled with relatively high winter SST and less than two months per year

of sea ice cover (Fig. 5b), indicating continued stratification. The isotopes of Npl suggest an influx of a warmer, more ventilated subsurface water mass. The timing of this shift roughly coincides with a retreat in the Hudson Strait Ice Stream (Dyke and Prest, 1987) between the Gold Cove Advance (~ 11.4 – 11.0 cal ka BP; Andrews et al., 1999) and the Nobel Inlet Advance (~ 10.2 – 9.5 cal ka BP; Jennings et al., 1998). Nearly simultaneously, the Northwest Passage became fully deglaciated (by ~ 10.3 ka; cf. Dyke, 2004) and Nares Strait began to open (~ 10.6 cal ka BP; England et al., 2006). The deglaciation of northeastern Canada would have been fostered by climate warming due to maximum summer insolation (Berger and Loutre, 1991). Therefore ~ 10.4 cal ka BP marks an interval of increased stratification due to intense meltwater input from the LIS and Arctic channels, and possibly a high subsurface inflow of NEADW in response to its enhanced overflow from the Nordic Seas during the early Holocene (e.g., Weaver and Hillaire-Marcel, 2004; see also Levac et al., 2001; Lloyd et al., 2005; Knudsen et al., 2008). The warmer sea surface conditions reflect the maximum warming of the early Holocene Thermal Optimum observed in other records from the Northwest Atlantic (de Vernal and Hillaire-Marcel, 2006).

5.3. The postglacial phase (~ 8.3 cal ka BP to present)

The averaged reconstruction for the ~ 8.3 cal ka BP layer assemblages suggests both warm winter and summer SSTs (Fig. 5b). Four of the 9 samples comprising the 36–85 cm interval contain specimens of the warm water species *S. mirabilis* (cf. Rochon et al., 1999). Assuming these specimens are not reworked, the assemblage might indicate some advection of North Atlantic waters to the northern Labrador Sea. Mild sea surface conditions, along with the negligible changes in the subsurface isotope values, correlate with other records from the Labrador Sea (Hillaire-Marcel et al., 2007) indicating a barely visible impact of the final drainage of glacial Lake Agassiz on surface and subsurface waters.

After ~ 8.3 cal ka BP, the dinocyst assemblages were dominated by *O. centrocarpum* ($>60\%$) with few remaining heterotrophic taxa, which suggest relatively warm winter and cool summer SSTs and thus a reduced seasonal temperature gradient (Figs. 4 and 5b; Rochon et al., 1999). They also indicate increased salinity, and decreased sea ice extent. Such sea surface conditions are attributed to reduced meltwater from the breakup of the LIS, which had retreated by more than 90% by ~ 7.5 cal ka BP (Dyke, 2004). Less meltwater resulted in reduced stratification thus giving way to winter convection and ventilation of the intermediate water mass, as shown by the switch to high $\delta^{13}\text{C}$ values in the subsurface dwelling Npl. This regime shift may be associated with inception of intermediate LSW production, as initially documented in core P094 at Orphan Knoll (Hillaire-Marcel et al., 2001) upon the demise of the LIS. A strengthened Irminger Current (IC) as it was suggested from dinocyst data in the eastern Labrador Sea (P013, Fig. 1; Solignac et al., 2004, 2006) would have contributed to enhanced supply of saltier water to the western Labrador Sea and promoted winter convection. This was the second major postglacial reorganization of surface and subsurface conditions in the northwest Labrador Sea.

The Holocene record of PC04 is characterized by a poor temporal resolution between ca 8 and 4 cal ka BP, preventing detailed interpretation of regional changes for this interval. The upper part of the sequence spanning the last 4000 years has a better resolution and permits some interpretations about the regional paleoceanography. Sea surface conditions were relatively warm at ca 4–3 cal ka BP (Fig. 5a), which coincides with the Holocene Thermal Maximum in terrestrial climate records described by Kaufman et al. (2004) and with the northward most migration of marine mollusks

along Baffin Island coastlines reported much earlier by Andrews (1972). After ~2.0 cal ka BP, the dinocyst-based reconstructions from core PC04 indicate colder, less saline sea surface conditions with increased ice cover (Fig. 5a). The ^{13}C -depleted and ^{18}O -enriched Npl suggest more stratified subsurface waters, possibly related to a weakened IC. These conditions could have developed as the results of enhanced flux of cooler, fresher Arctic surface water from the Canadian Arctic Archipelago (CAA) to the Labrador Current. Levac et al. (2001) associated similar sea surface conditions in northern Baffin Bay to an increase in Arctic water flux through Nares Strait. Jennings et al. (2011) also documented an increase in Polar Water in the EGC relative to a decrease in the contribution of Irminger water. Therefore, the recent trend towards colder, less saline surface waters in the northwest Labrador Sea suggests that there is an increase in Arctic outflow with a strengthened Baffin Island Current and weakened IC, coinciding with the Neoglacial cooling of the northeast Canadian coast (Miller et al., 2005).

6. Conclusions

Palynological analyses and foraminiferal isotopic analyses of a core from the outlet of Hudson Strait in the deep northwest Labrador Sea revealed that the regional hydrography has been strongly influenced by meltwater discharge from the LIS and/or freshwater fluxes from the Arctic. Several features are singular to the site. Firstly, proximal LIS margin conditions were maintained until ~12.2 cal ka BP. This glacial phase was characterized by quasi-perennial sea ice conditions, with episodes of sea ice break up and extreme sea ice formation at the termination of Heinrich Events. Secondly, both the BA and YD climatic events were not clearly recorded. Thirdly, during the local deglacial interval from ca 12.2 to 8.3 cal ka BP, meltwater from the retreating LIS contributed to low sea surface salinity, as low as 28 in the surface water, which resulted in a strong stratification that prevented convective mixing. This allowed a strong summer warming with SST reaching up to 11 °C and reduced sea ice cover (down to 3 months per year). As a consequence, the habitat of the planktic foraminifer Npl, which requires salinity above 34 to develop (e.g., Spindler and Dieckmann, 1986; Bergami et al., 2009), was displaced towards the deeper end of the pycnocline with the intermediate water mass where such conditions could be found. This behavior would thus account for the departure of isotopic records in planktics from values in agreement with temperatures and salinities of the surface water layer recorded by dinocysts. Finally, one may interpret the complicated paleoceanography of the area as a response to the combined influences of the thermal conditions in the North Atlantic, the local ice history and the late opening of the Canadian Arctic channels. Therefore, the major paleoceanographic reorganizations observed in the northwest Labrador Sea at ca 12.2 and 8.3 cal ka BP are regionally consistent but have no analogue elsewhere. This again highlights the strong regionalism characterizing the North Atlantic oceanographic response to climatic changes (Solignac et al., 2004, 2006; de Vernal and Hillaire-Marcel, 2006).

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